

# BRAF / MEK + PD-(L)1: 2 molecules stopped early, all small molecules

3 stopped trials, but 2 molecules — all small molecules: Cobimetinib and Binimetinib. Against 4.9% for PD-(L)1 combinations without BRAF / MEK and 4.3% across all oncology Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

4.7%

3 of 64 closed trials (95% CI 1.6%–12.9%)

1 / 0 / 1

efficacy / safety / both (adds to 3)

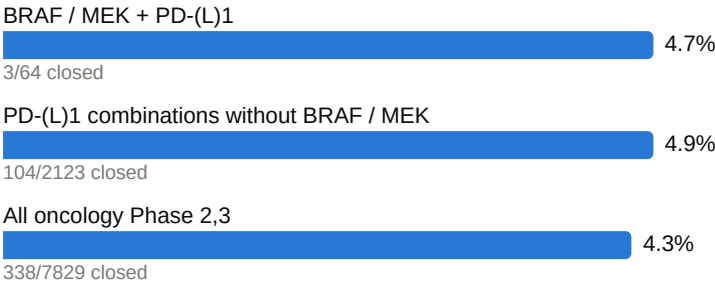
39.8%

of 161 trials have closed · 97 still open or unresolved

1.6%

stopped for a biological reason within 3 years of starting (95% CI 0.4%–6.4%; 97 still at risk; 1.5% area-wide)

## How this compares



**What is in this class:** 190 distinct experimental drugs across 161 trials (small molecule 161, antibody 151, unknown 27, adc 12). Most tested: REGORAFENIB (46), COBIMETINIB (42), ATEZOLIZUMAB (42), NIVOLUMAB (32), TRAMETINIB (23). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 3 stops came from 3 sponsor-asset programmes across 3 sponsors; the largest (Roche) contributed 1; removing that programme's trials from both sides leaves 3.3%. Screened alongside 85 other segments in this area, it does not survive a 10% false-discovery correction (q=1, Benjamini-Yekutieli). A further 4 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 10.9%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI     |
|----------------|-------|--------|------|------------|
| 2015–2017      | 1     | 22     | 4.5% | 0.8%–21.8% |
| 2018–2020      | 2     | 25     | 8.0% | 2.2%–25.0% |
| 2021–2024      | 0     | 17     | 0.0% | 0.0%–18.4% |

## Where the stops sit

Efficacy and safety: 1, Efficacy / futility: 1, Benefit–risk: 1. Sponsors with most stops: Columbia University (1), Pfizer (1), Roche (1).

## The stopped trials

| Trial       | Ph | Sponsor                 | Experimental drugs                            | Stopped | Type              | Registry stop reason                                                                                                          |
|-------------|----|-------------------------|-----------------------------------------------|---------|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| NCT04214418 | 2  | Columbia University     | COBIMETINIB, HYDROXYCHLOROQUINE, ATEZOLIZUMAB | 2022-04 | Efficacy + safety | Toxicity and Lack of Efficacy                                                                                                 |
| NCT03637491 | 2  | Pfizer <span>PFE</span> | AVELUMAB, BINIMETINIB, TALAZOPARIB            | 2020-12 | Efficacy          | Available clinical data has shown limited anti-tumor activity and reaching target study drug dose levels may not be feasible. |
| NCT03273153 | 3  | Roche                   | COBIMETINIB, ATEZOLIZUMAB                     | 2019-04 | Benefit–risk      | This study was terminated due to benefit/risk analysis.                                                                       |

### Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated).  
Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

### Limits

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 39.8% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**Any mechanism, sponsor or indication, updated weekly.** The dataset behind this brief covers every stopped oncology trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: [clinicaltrialfailures.com/pricing](https://clinicaltrialfailures.com/pricing)

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.